Recursion and Memoization

Problem Set

Problem 1 — Colored Tiling of a Strip

You want to completely tile a $1 \times N$ strip using two kinds of tiles:

- a 1×1 tile, available in 3 distinct colours, and
- a 1×2 tile, available in only 1 colour.

Two tilings are considered different if, at some position, a different tile (size or colour) is used.

Find the number of distinct tilings of a $1 \times N$ strip, for the following values of N:

(a) $N = 4$

(b) $N = 5$

(c) $N = 6$

Problem 2 — Binary Clock

A binary string of length 12 is a sequence consisting only of the symbols 0 and 1.

A binary string is called **good** if it contains neither

$$000$$

nor

$$111$$

as a consecutive substring.

For example, the string 0010110 is good, while 0011101 is not.

How many good binary strings of length 12 are there?

Problem 3 — Weighted Ternary Strings

Consider ternary strings formed using digits from the set $\{0, 1, 2\}$.

The *weight* of a string is defined as the sum of its digits, where digits 0 and 1 each contribute a weight of 1, while the digit 2 contributes a weight of 2.

A string is called *valid* if it contains no consecutive 2s (that is, the substring 22 does not appear anywhere in the string).

Find the total number of valid ternary strings of total weight N in the following cases:

(a) $N = 5$

(b) $N = 7$

(c) $N = 10$

Problem 4 — Restricted Distance Jumps

A token is placed at position 0 on a line. In each step, the token can jump forward by a distance of 1, 2, or 4 units. A sequence of jumps is valid if it starts at position 0, ends at position N, and satisfies the following conditions for every pair of consecutive jumps:

1. A jump of size 2 cannot immediately follow a jump of size 1.
2. A jump of size 4 cannot immediately follow a jump of size 2.

The first jump made from position 0 has no preceding jump restrictions and can be of size 1, 2, or 4.

Compute the total number of valid jump sequences that end exactly at position N in the following cases:

(a) $N = 7$

(b) $N = 10$

(c) $N = 15$